

MOHAMED YASSINE KOUBAA

Robotics & Embedded System Engineering

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ABOUT

Passionate robotics engineer with hands-on experience in embedded system. Adept in AI-driven computer vision solution and developing innovative systems for real-world application. Currently holding position as Software developer at Telnet Holding and eager to leverage my expertise in Python, C/C++ in impactful projects. Looking to drive technological advancements and contribute to groundbreaking developments in the tech industry.

SKILLS

Programming Languages	C/C++, Embedded C, Python, JavaScript(NodeJS)
Frameworks & Libraries	OpenCV, TensorFlow, PyTorch, ExpressJS, ReactJS
Robotics	ROS, Localization, Autonomous Navigation, Path Planning, Sensor Fusion, Control
SW Development Practices	Git/Gitlab/Github, CI/CD, Scrum, JIRA
Soft Skills	Time Management, Teamwork, Communication, Problem Solving, NLP.

EXPERIENCE

C/C++ Developer (Worldline Partner) Jun 2023 - Present
Telnet Holding *Sfax, Tunisia*

- Developing and maintaining C/C++ drivers for payment terminals, enabling secure POS-to-terminal communication.
- Implementing interfaces for terminal application, enabling seamless HMI and server/POS communication integration.
- Enhancing Terminal Management System (TMS) capabilities allowing remote update and configuration.
- Ensuring code quality through reviews and documentation while maintaining compliance with security standards.

Keywords: C/C++, CMake, Secure TCP/IP communication, Serial Communication, Git/Gitlab, CI/CD.

Robotics Research Intern Feb 2022 - Jul 2022
Hydrogen Research Institute (UQTR) *Trois Rivières, QC, Canada*

- Researched state-of-the-art 6-DOF vision-based object localization techniques, identifying gaps and improvements.
- Designed and trained an AI-driven novel object localization solution with an augmented auto-encoder, using PyTorch.
- Developed a position refinement algorithm using 3D contours from rendered depth images and template matching techniques, enhancing localization accuracy.
- Integrated and tested the solution within the AGV's ROS ecosystem, ensuring reliability in a simulated environment.

Keywords: Artificial Intelligence, CNN, Auto-Encoder, PyTorch, OpenCV, Transform Matrix, ROS, Gazebo, Linux.

Computer Vision Summer Intern Jul 2021 - Aug 2021
R2A Solutions *Tunis, Tunisia*

- Built a background subtraction-based object detection algorithm with OpenCV for faster YOLOv3 dataset annotation.
- Created a Qt interface for dataset creation and generation, using OpenCV and Qt Concurrent for speed and reactivity.
- Automated dataset creation in ROS, using a conveyor and delta robot to place objects in the FOV for efficient collection.
- Trained a YOLOv3 model with transfer learning on the generated dataset, achieving high accuracy in real-case testing.

Keywords: C/C++, OpenCV, Qt, ROS, Linux, CNN, DarkNet, YOLOv3.

Embedded System Summer Intern Jul 2020
Sofia Technologies *Ariana, Tunisia*

- Designed a drone gesture controller using an IMU with a complementary filter for noise-free recognition.
- Programmed BLE on NRF52 for low-latency sensor data transfer and reliable communication between nodes.
- Built a differential-drive robot controlled by pitch and yaw, showcasing the gesture controller's precision and reactivity.

Keywords: Embedded C, Nordic NRF52, BLE, I2C, SPI.

EDUCATION

M2 Research Master Degree in Information System Techniques Sept 2021 - Dec 2022
National Engineering School of Tunis

Engineering Degree in Industrial Computing and Automation Sept 2017 - Oct 2022
National Institute of Applied Science and Technology

National Baccalaureate Diploma in Technical Sciences Sept 2013 - Jun 2017
El Habib High School

PROJECTS

Autonomous Robots: EuroBot 2019

Video

- Built 2 autonomous robots with STM32F4, ATmega328P, encoders, proximity sensors, beacons, DC motors, and pumps.
- Implemented PID control with encoder feedback, ensuring precise, repeatable velocity and position control.
- Designed a beacon-based avoidance system with hardware integration and a custom algorithm for obstacle detection.
- Enhanced system reliability via rigorous testing, ensuring consistent performance in dynamic settings.
- Contributed to hardware and software integration, including sensors, actuators, and real-time controls.
- Secured 2nd place in Tunisian qualifiers and 7th at Eurobot finals in Roche-sur-Yon, France.

Keywords: Embedded C, C++, STM32, ATmega, Control, PID, Path Planning, CAD, SolidWorks, Proteus.

Streaming Overlay Platform: Eurobot Score Stream Overlay

GitHub

- Developed a scoring platform with ReactJS, Express.js, and MongoDB for centralizing match data input and storage.
- Designed an AV dashboard with predefined overlays, allowing streamers to manage live visuals in OBS.
- Implemented real-time overlays using Socket.IO, displaying scores, countdowns, and round details with animations.
- Built JWT-based authentication with four user roles (visitor, streamer, referee, admin) for secure access.
- Deployed using Docker, ensuring easy setup and platform customization for each competition edition.
- Received positive feedback from viewers and referees for the platform's professionalism and ease of use.

Keywords: ReactJS, ExpressJS, MongoDB, Socket.IO, JWT Authentication, User Roles, Docker, OBS.

Motion Robotics Framework:

Video

- Architected a C++ abstraction layer decoupling hardware logic from high-level algorithms for cross-platform portability.
- Designed "Motion Core" a robust nucleus for real-time odometry and sensor fusion using modern C++ patterns.
- Developed "Motion Link" middleware, abstracting communication layers to enable telemetry and command injection.
- Implemented a multi-threaded RTOS architecture using mutexes and queues for thread-safe task execution.
- Designed a flexible API for rapid prototyping with low-level hooks for fine-grained robot control.
- Authored comprehensive documentation and integration examples, significantly reducing developer onboarding and system setup time.

Keywords: : C/C++, Embedded Systems, RTOS, Multi-Threading, Modular Architecture, Abstraction Design Patterns, Protocol Management, Real-Time Systems.

CERTIFICATION

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| • Certified SolidWorks Associate (CSWA) | Jan 2019 |
| • Neuro Linguistic Programming Practitioner | Aug 2020 |
| • Certified SolidWorks Professional (CSWP) | Oct 2020 |

LANGUAGES

- **Arabic:** Native
- **French:** Advanced
- **English:** Advanced

SOCIAL ACTIVITIES

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| • AeRobotiX INSAT: Technical Vice-President | Jul 2020 - Jul 2021 |
| • ATR: Board Member | Oct 2018 - Mar 2023 |
| • BIAT Youth Foundation: Generation Leaders Elite Program | Sep 2017 - Jun 2022 |
| • RobOlympiX: Event Founder | Dec 2020 |
| • AeroDay: Volunteer | Dec 2017 - Oct 2021 |